

VERESHCHAGINO PERM KRAI, Russian Federation

WMO: 282160

Lat: 58.083N

Lon: 54.683E

Elev: 226

StdP: 98.63

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.5	-26.3	-32.3	0.2	-29.2	-29.1	0.3	-26.0	9.5	-11.2	8.3	-10.3	1.0	0	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.1	20.1	27.4	19.5	25.6	18.6	21.7	27.3	20.5	25.9	19.3	24.3	2.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.9	25.2	18.5	13.7	23.6	17.2	12.7	22.3	64.5	27.2	59.9	26.2	55.9	24.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.8	5.5	-32.4	31.5	3.8	2.0	-35.1	32.9	-37.3	34.1	-39.4	35.2	-42.2	36.7		
(5)			-32.8	23.7	3.8	1.6	-35.6	24.8	-37.8	25.8	-40.0	26.7	-42.8	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-13.4	-11.8	-4.4	3.7	11.2	16.3	18.8	15.8	9.9	2.7	-5.4	-11.2	(6)
(7)		DBStd	12.53	7.24	6.83	4.37	5.12	5.37	4.35	4.32	4.36	4.45	4.68	6.36	7.20	(7)
(8)		HDD10.0	3452	726	612	446	198	52	4	1	4	56	233	462	658	(8)
(9)		HDD18.3	5809	984	845	704	438	226	88	48	105	256	486	712	917	(9)
(10)		CDD10.0	807	0	0	0	9	91	194	273	182	51	6	0	0	(10)
(11)		CDD18.3	123	0	0	0	0	6	28	62	26	2	0	0	0	(11)
(12)		CDH23.3	963	0	0	0	0	51	216	501	186	9	0	0	0	(12)
(13)		CDH26.7	231	0	0	0	0	6	41	134	49	1	0	0	0	(13)
(14)	Wind	WSAvg	2.2	2.3	2.3	2.4	2.1	2.3	2.1	1.6	1.6	1.8	2.3	2.4	2.5	(14)
(15)	Precipitation	PrecAvg	592	40	26	39	34	44	67	68	67	58	54	50	42	(15)
(16)		PrecMax	803	74	41	118	76	95	165	163	163	137	104	83	73	(16)
(17)		PrecMin	407	8	9	3	1	12	19	8	8	10	10	14	15	(17)
(18)		PrecStd	88	16	10	23	21	20	33	35	34	29	24	18	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	1.1	8.4	21.7	27.7	29.6	31.7	31.4	24.7	17.3	8.4	2.2	(19)
(20)			MCWB	0.1	-0.1	4.7	13.4	16.5	20.1	21.1	20.7	18.5	12.1	7.3	1.7	(20)
(21)		2%	DB	-0.5	0.1	4.6	17.4	24.7	27.8	29.7	27.4	21.1	13.9	5.2	0.6	(21)
(22)			MCWB	-1.0	-0.7	1.7	11.2	16.0	19.1	20.6	19.9	15.7	10.1	4.4	0.0	(22)
(23)		5%	DB	-1.9	-0.9	2.7	14.2	22.6	25.9	28.2	25.1	18.7	11.3	3.0	-0.4	(23)
(24)			MCWB	-2.5	-1.6	0.4	9.1	14.7	18.3	20.2	18.7	14.5	8.8	2.3	-0.9	(24)
(25)		10%	DB	-3.6	-2.5	1.2	11.4	20.2	23.9	26.3	22.8	16.5	9.1	1.2	-2.1	(25)
(26)			MCWB	-4.2	-3.2	-0.4	7.0	13.6	17.3	19.5	17.8	13.2	7.5	0.5	-2.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.3	5.0	14.4	18.9	22.3	23.6	22.3	18.7	12.7	7.3	1.7	(27)
(28)			MCDWB	0.3	0.7	7.9	19.9	25.0	27.5	29.0	28.1	23.7	16.4	7.9	2.0	(28)
(29)		2%	WB	-0.9	-0.5	2.2	11.6	16.8	20.6	22.2	20.7	16.6	10.5	4.6	0.1	(29)
(30)			MCDWB	-0.5	0.0	4.0	16.9	23.1	26.0	27.8	26.6	20.5	13.3	5.3	0.4	(30)
(31)		5%	WB	-2.5	-1.6	0.9	9.4	15.5	19.2	21.1	19.3	14.9	9.0	2.4	-0.9	(31)
(32)			MCDWB	-2.0	-0.9	2.0	14.0	21.7	24.3	26.5	24.2	18.1	10.8	3.0	-0.4	(32)
(33)		10%	WB	-4.1	-3.2	-0.2	7.4	14.1	17.9	19.9	18.0	13.5	7.6	0.6	-2.6	(33)
(34)			MCDWB	-3.6	-2.5	0.9	10.9	19.2	22.6	25.0	22.0	16.0	9.1	1.1	-2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	6.2	6.7	8.1	9.7	9.1	9.2	8.3	7.0	4.8	4.1	5.1	(35)
(36)			MCDBR	5.5	5.5	8.0	11.9	12.7	11.4	11.6	11.0	9.8	8.0	4.5	4.5	(36)
(37)			MCWBR	5.4	5.0	6.0	7.8	7.2	6.4	6.3	5.7	5.9	5.6	4.3	4.5	(37)
(38)		5% WB	MCDWR	5.5	5.2	6.3	11.1	11.6	10.8	10.9	10.3	9.1	7.3	4.2	4.4	(38)
(39)			MCWBR	5.4	4.9	5.3	8.0	7.4	6.8	7.0	6.2	6.0	5.5	4.2	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.281	0.310	0.361	0.389	0.381	0.377	0.397	0.411	0.367	0.338	0.299	0.274	(40)	
(41)		taud	2.014	2.031	1.960	2.101	2.329	2.386	2.335	2.278	2.370	2.303	2.124	1.983	(41)	
(42)		Ebn at Noon	516	690	753	810	852	861	830	786	768	671	500	393	(42)	
(43)		Edh at Noon	57	90	130	133	114	110	113	112	87	70	51	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.52	1.40	2.72	4.03	5.12	5.48	5.48	4.10	2.40	1.05	0.42	0.28	(44)	
(45)		RadStd	0.08	0.24	0.33	0.52	0.57	0.49	0.58	0.37	0.28	0.14	0.08	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page